DEVELOPER ROLE

SET D — 100 MCQs

Practice assessment · Java, DSA, SQL/DBMS, web, Spring Boot, security, Git/testing, OS and system design.

Instructions: Select one best answer per question. Suggested time: 75 minutes. One mark per question; no negative marking in this practice version.

1. Which ACID property means all-or-nothing?

A. Isolation
B. Durability
C. Consistency
D. Atomicity

2. Which interface is the root of the Java collection hierarchy for groups of elements?

A. Collection
B. Comparable
C. Runnable
D. Serializable

3. Insert at the head of a linked list when head is known takes?

A. O(n)
B. O(log n)
C. $O(n^2)$
D. O(1)

4. Why paginate API results?

A. Encrypt payloads
B. Prevent all errors
C. Replace authorization
D. Return large result sets in manageable chunks

5. Which structure is commonly used for efficient prefix searches?

A. Binary heap
B. Trie
C. Stack
D. Queue

6. What is method overloading?

A. Same name with different parameter lists
B. Hiding a field
C. Subclass replacement of a method
D. Changing a return value only

7. What does CORS govern?

A. Database migrations
B. Password hashing
C. Cross-origin browser requests
D. CPU scheduling

8. What is the purpose of a database index?

A. Speed up selected reads at storage/write trade-off
B. Encrypt every row
C. Prevent all duplicates automatically
D. Replace primary keys

9. Average lookup in a well-distributed hash table is typically?

A. O(log n)
B. O(1)
C. O(n log n)
D. O(n)

10. Which SQL command retrieves rows?

A. SELECT
B. OPEN
C. FETCH TABLE
D. GET

11. What is the purpose of memoization?

A. Cache results of repeated function calls
B. Sort linked lists
C. Increase recursion depth
D. Encrypt variables

12. Which data structure follows LIFO?

A. Stack
B. Graph
C. Heap
D. Queue

13. Which Java collection rejects duplicate elements?

A. LinkedList
B. HashSet
C. ArrayList
D. Vector

14. What does GROUP BY do?

A. Deletes duplicates
B. Groups rows for aggregation
C. Sort alphabetically
D. Renames a table

15. What does HTTPS add to HTTP?

A. Guaranteed zero latency
B. Automatic database caching
C. TLS transport security
D. Mandatory compression

16. Which HTTP method is generally used to retrieve a resource?

A. PATCH
B. DELETE
C. POST
D. GET

17. Which data structure follows FIFO?

A. Stack
B. Set
C. Queue
D. Tree

18. Normalization primarily reduces what?

A. Redundancy and update anomalies
B. Indexes
C. All joins
D. Query syntax

19. Which structure supports the call stack used by recursion?

A. Hash table
B. Stack
C. Disjoint set
D. Queue

20. What is Git primarily used for?

A. Running production servers
B. Relational storage
C. Version control and collaboration
D. Web design

21. Which standard Java class is immutable?

A. StringBuilder
B. StringBuffer
C. ArrayList
D. String

22. Which BST traversal yields sorted keys?

A. Preorder
B. Postorder
C. Reverse level order
D. Inorder

23. What does COUNT(*) count?

A. Only non-null values in one column
B. Only distinct rows
C. Only primary keys
D. Rows in the result set

24. What does a foreign key enforce?

A. Encryption
B. Sorting
C. Compression
D. Referential integrity

25. Which data structure is suitable for union/find connectivity queries?

A. Disjoint-set union
B. Deque only
C. Stack
D. Trie

26. What does COMMIT do?

A. Undo a transaction
B. Delete a database
C. Create an index
D. Persist transaction changes

27. What does DNS do?

A. Compile Java
B. Resolve domain names to IP addresses
C. Create database keys
D. Encrypt HTTP traffic

28. What does static mean for a class field?

A. The field is always immutable
B. A class-level member shared by instances
C. One copy per object
D. The field cannot be read

29. What is a database transaction?

A. File format
B. Index type
C. User account
D. Logical unit of work committed or rolled back

30. Which access modifier gives the broadest visibility?

A. public
B. package-private
C. private
D. protected

31. Which Java keyword prevents class inheritance?

A. private
B. volatile
C. final
D. static

32. Which OOP principle hides internal state behind methods?

A. Inheritance
B. Abstraction
C. Polymorphism
D. Encapsulation

33. What does git pull typically do?

A. Creates a hosted repository
B. Deletes local branches
C. Rewrites all commit messages
D. Fetches and integrates remote changes

34. Which status code represents an internal server error?

A. 201
B. 500
C. 404
D. 302

35. What does a repository usually abstract?

A. Browser history
B. CSS rendering
C. Data access operations
D. DNS configuration

36. Which technique explores a graph level by level?

A. BFS
B. Heapify
C. DFS
D. Backtracking only

37. What does @Transactional help manage?

A. Network routes
B. Database transaction boundaries
C. HTML layout
D. JWT keys only

38. Which media type represents JSON?

A. application/json
B. image/java
C. application/java
D. text/sql

39. What is a DTO used for?

A. Transfer data across application boundaries
B. Start threads
C. Replace every service
D. Exact database engine representation

40. Which HTTP method is commonly used for partial updates?

A. PATCH
B. TRACE
C. OPTIONS
D. GET

41. A balanced BST typically supports search in what time?

A. O(2■)
B. O(n^2)
C. O(1) guaranteed
D. O(log n)

42. Which HTTP status commonly means resource created?

A. 201
B. 404
C. 200
D. 301

43. Binary search on a sorted array takes what worst-case time?

A. O(1)
B. O(log n)
C. O(n^2)
D. O(n)

44. Which SQL statement changes existing rows?

A. UPDATE
B. GRANT
C. ALTER
D. CREATE

45. What is true about Java interfaces?

A. A class can extend multiple classes
B. Interfaces cannot have default methods
C. Interfaces require constructors
D. A class can implement multiple interfaces

46. Which keyword makes a variable's visibility changes observable across threads but does not make compound operations atomic?

A. strictfp
B. transient
C. volatile
D. native

47. Which clause filters aggregated groups?

A. LIMIT
B. WHERE
C. DISTINCT
D. HAVING

48. What is the default value of an instance int field in Java?

A. 0
B. -1
C. null
D. Uninitialized random value

49. Which algorithm finds shortest paths in an unweighted graph?

A. BFS
B. Binary search
C. Bubble sort
D. DFS

50. Which Spring annotation marks a REST controller?

A. @ServletTable
B. @AutowiredOnly
C. @RestController
D. @EntityManager

51. Which keyword explicitly throws an exception?

A. throws
B. final
C. catch
D. throw

52. Dynamic programming is especially useful when subproblems are what?

A. Overlapping with reusable results
B. Limited to graphs
C. Always independent
D. Only sorted

53. Where should production secrets be kept?

A. Committed source code
B. Frontend bundle
C. Secret manager/protected environment
D. Public README

54. Which structure efficiently retrieves the highest-priority item?

A. Stack
B. Basic queue
C. Singly linked list
D. Priority queue

55. Which layer should normally handle HTTP request mapping?

A. Repository
B. Build tool
C. Controller
D. Database index

56. What is a constructor used for?

A. Handling exceptions only
B. Starting a thread automatically
C. Initializing a new object
D. Destroying an object

57. What does a generic type parameter provide?

A. Compile-time type safety
B. Runtime encryption
C. Thread scheduling
D. Automatic persistence

58. How should passwords be stored?

A. Reversible Base64
B. Plain text
C. Unsalted MD5
D. Salted slow hash such as BCrypt/Argon2

59. Which block commonly runs whether or not an exception occurs?

A. throw
B. catch only
C. assert
D. finally

60. Which is a checked Java exception?

A. ArrayIndexOutOfBoundsException
B. ArithmeticException
C. NullPointerException
D. FileNotFoundException

61. Which Java type stores key-value pairs?

A. ArrayList
B. PriorityQueue
C. HashMap
D. HashSet

62. REST is best described as what?

A. Architectural style for networked applications
B. Programming language
C. Database engine
D. CSS framework

63. Which algorithmic strategy chooses a locally optimal option at each step?

A. Hashing
B. Normalization
C. Dynamic linking
D. Greedy

64. What helps prevent SQL injection?

A. Exposing SQL errors
B. Disabling indexes
C. Concatenating user input into SQL
D. Parameterized queries

65. When properly overridden, equals() usually checks what?

A. Reference identity only
B. Object size
C. Class loading order
D. Logical equality

66. What is Flyway commonly used for?

A. UI icons
B. Load balancing
C. Versioning and applying database schema changes
D. Bytecode compression

67. Which key uniquely identifies a table row?

A. Foreign key
B. Primary key
C. Duplicate key
D. Secondary index

68. Which collection preserves insertion order and allows duplicates?

A. ArrayList
B. HashSet
C. TreeSet
D. HashMap

69. What does least privilege mean?

A. Everyone gets admin access
B. Grant only necessary permissions
C. Share one account
D. Disable authentication

70. An adjacency-list graph representation typically uses what space?

A. O(V + E)
B. $O(E^2)$
C. O(log V)
D. $O(V^2)$ always

71. Which algorithm finds a minimum spanning tree?

A. KMP
B. Floyd cycle detection
C. Kruskal's algorithm
D. Binary search

72. What is the typical average time complexity of hash-table insertion?

A. $O(n^2)$
B. O(log n)
C. O(n log n)
D. O(1)

73. Which sorting algorithm is typically stable?

A. Merge sort
B. Heap sort
C. In-place quicksort
D. Selection sort

74. What is quicksort's worst-case time complexity?

A. O(log n)
B. O(n)
C. O(n log n)
D. $O(n^2)$

75. Which command removes all rows while retaining table structure?

A. RENAME
B. TRUNCATE
C. DROP
D. ALTER

76. Which join returns only matching rows from both tables?

A. INNER JOIN
B. CROSS JOIN
C. LEFT JOIN
D. FULL OUTER JOIN

77. Which index structure is common for database range queries?

A. B-tree/B+ tree
B. Stack
C. Call stack
D. Binary heap

78. Which command creates and switches to a branch?

A. git stash pop
B. git switch -c branch-name
C. git reset --hard
D. git fetch --delete

79. Which Java type is designed for mutable strings without synchronization?

A. String
B. StringLiteral
C. Character
D. StringBuilder

80. Which Spring annotation maps GET requests?

A. @DeleteMapping
B. @PutMapping
C. @PostMapping
D. @GetMapping

81. Which clause filters rows before grouping?

A. ORDER BY
B. WHERE
C. HAVING
D. GROUP BY

82. What is the time complexity of merging two sorted arrays of total length n?

A. O(1)
B. O(n)
C. $O(n^2)$
D. O(log n)

83. Which application layer usually contains business rules?

A. DNS
B. Presentation
C. Physical network
D. Service/business

84. Which constraint prevents NULL values in a column?

A. DEFAULT
B. NOT NULL
C. CHECK
D. FOREIGN KEY

85. Which file commonly holds Spring Boot configuration?

A. index.css
B. application.properties
C. routes.sql
D. pom.lock

86. Which SQL operator checks whether a value matches any value in a list?

A. IN
B. LIKE ONLY
C. IS NULL
D. BETWEEN

87. Which technique is commonly used to find connected components?

A. Hashing strings only
B. Binary search only
C. Graph traversal
D. Insertion sort

88. What does JVM stand for?

A. Joint Version Model
B. Java Variable Manager
C. Java Virtual Machine
D. Java Verified Module

89. Which HTTP status means authentication is missing or invalid?

A. 500
B. 401
C. 400
D. 403

90. What is SQL injection?

A. Backup method
B. Query optimization
C. Normalization form
D. Attack manipulating SQL via untrusted input

91. What is method overriding?

A. Calling a static method
B. Changing a variable type
C. Declaring multiple constructors
D. Subclass implementation of an inherited method

92. Which command changes a table definition?

A. UPDATE
B. COMMIT
C. SELECT
D. ALTER TABLE

93. What is the purpose of try-with-resources?

A. Suppress all exceptions
B. Retry failed requests
C. Create threads
D. Automatically close AutoCloseable resources

94. What is the Java application entry-point method?

A. run()
B. start()
C. public static void main(String[] args)
D. init()

95. What is dependency injection?

A. Encrypting method names
B. Creating every dependency inside a class
C. Converting Java to SQL
D. Supplying an object's dependencies externally

96. How can a linked-list cycle be detected with O(1) extra space?

A. Topological sort
B. Counting sort
C. Floyd slow/fast pointers
D. Binary search

97. Which keyword refers to the current Java object?

A. self
B. current
C. this
D. super

98. What does a JWT signature provide?

A. Confidentiality of all claims
B. Password storage
C. Infinite token lifetime
D. Integrity and authenticity verification

99. Which map implementation generally does not guarantee iteration order?

A. HashMap
B. TreeMap
C. LinkedHashMap
D. EnumMap

100. Which protocol provides reliable ordered byte-stream delivery?

A. ICMP
B. TCP
C. ARP
D. UDP